Retail Client Pipeline Options – 3 Tier Model

A clear, scalable architecture model for retail clients based on data volume, latency needs, and cost efficiency.

Tier 1: Low-Scale Clients (No Snowpipe Needed)

Ideal for: Small retailers, boutiques, cafes.
Data Volume: < 1 GB/day.
Latency Requirement: Daily or every few hours.

Pipeline Pattern:
• Lambda/ECS → S3 (raw zone)
• Scheduled COPY INTO Snowflake
• dbt run daily or twice per day

Benefits:
• Extremely low cost
• Simple to maintain
• No need for real-time ingestion

Tier 2: Medium-Scale Clients (Snowpipe Recommended)

Ideal for: Multi-store retailers, busy supermarkets.
Data Volume: 1–20 GB/day.
Latency Requirement: 15–60 minutes.

Pipeline Pattern:
• Lambda/ECS every 5–15 minutes → S3
• Snowpipe auto-ingest into Snowflake
• dbt runs every 15–30 minutes

Benefits:
• Automated ingestion
• Handles higher volume
• Supports intra-day decision-making

Tier 3: High-Scale Clients (Streaming + Near Real-Time)

Ideal for: Large chains, enterprise retail, high-velocity e-commerce.
Data Volume: 20 GB+/day.
Latency Requirement: 1–5 minutes (near real-time).

Pipeline Pattern:
• Kinesis/MSK streams → S3
• Snowpipe Streaming or micro-batch ingestion
• Snowflake Tasks + incremental dbt models

Benefits:
• Scales horizontally
• Near real-time dashboards
• Enterprise-grade performance

Quick Comparison Table

Tier	Volume	Latency	Ingestion	Best For
Low	<1 GB/day	Daily	Batch COPY	Small shops
Medium	1–20 GB/day	15–60 min	Snowpipe	Multi-store
High	20+ GB/day	1–5 min	Streaming	Enterprise

Use this model to match clients to the simplest pipeline that fits their data needs without over-engineering.